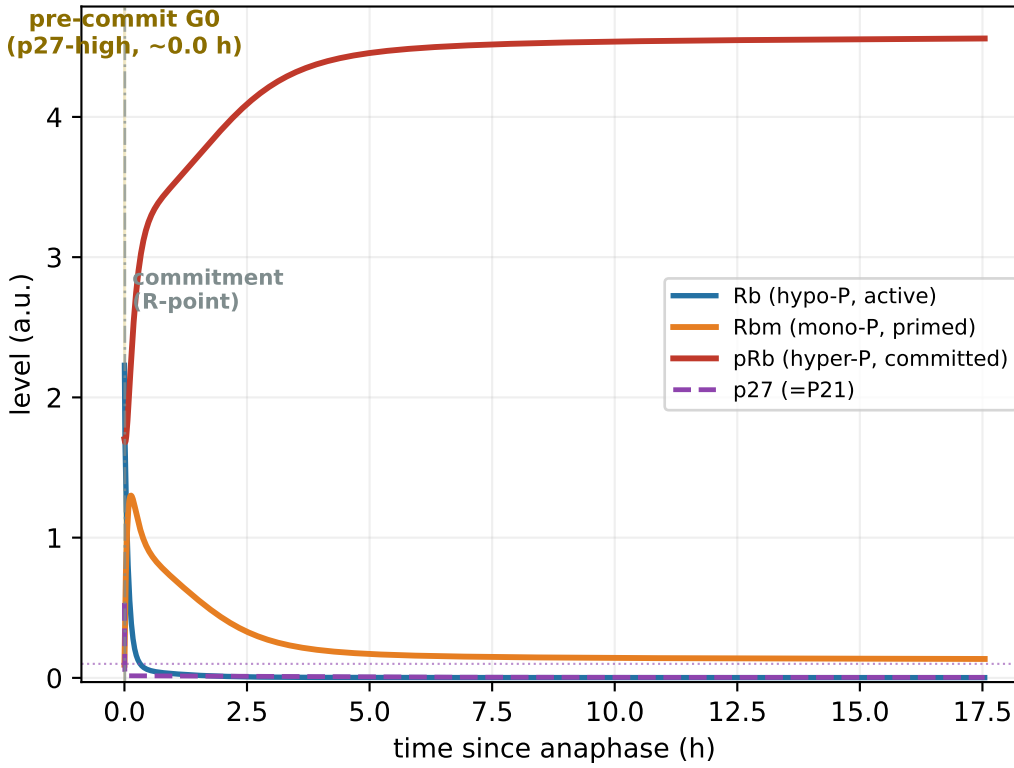


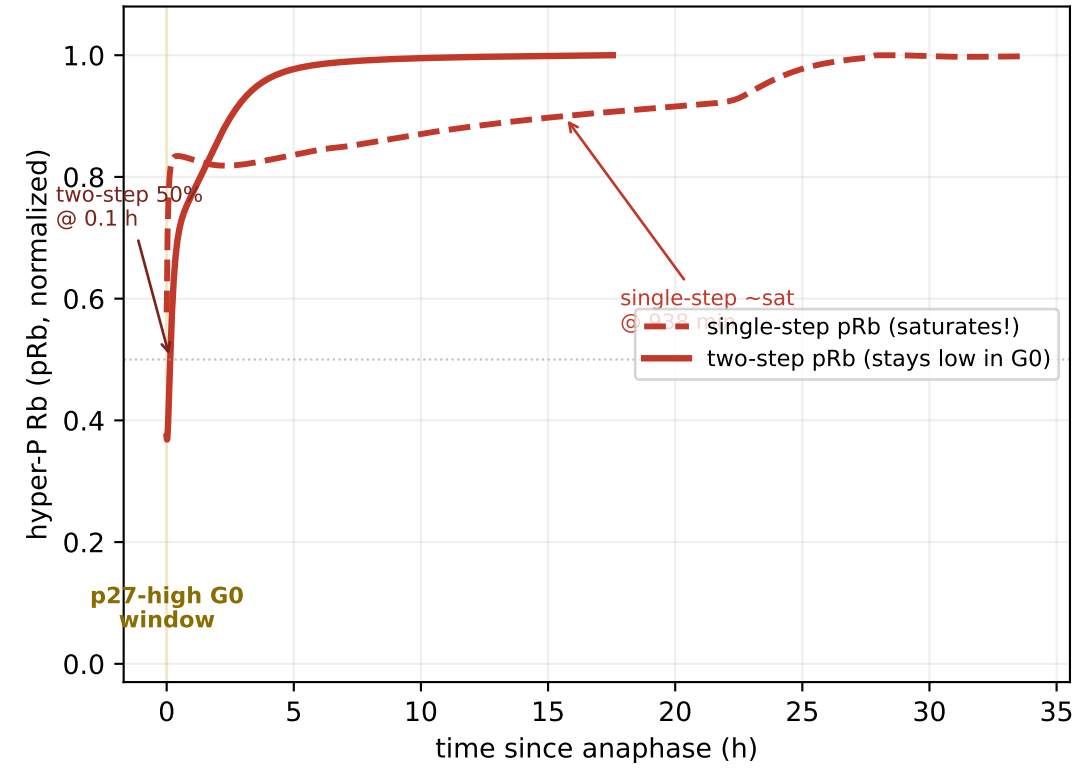
Two-step Rb switch restores hyper-P Rb as a faithful G0 / R-point marker (v44 default)

CyclinD-CDK4/6 primes Rb→Rbm (mono-P); CyclinE/A-CDK2 commits Rbm→pRb (hyper-P) at the R-point. GNP transient-G0 cell. Single-step pRb is high from anaphase (bad marker); two-step pRb stays low through G0 (good marker).

**(a) One GNP cycle (two-step default):
Rb(hypo) → Rbm(mono-P) → pRb(hyper-P)**



**(b) Hyper-P Rb across a cycle:
single-step saturates ~20 min post-anaphase vs two-step**



**(c) Marker agreement in the pre-commit G0 window:
p27-HIGH ↔ hyper-P-Rb-LOW**

